

Sara Miller

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Portfolio & Selected Projects [Sara Miller | Engineering Portfolio](#)

EDUCATION

BESc. Mechatronics Engineering University of Western Ontario, London Ontario 2022 - Expected May2027

- Granted distinction of Dean's List for high academic achievement

WORK EXPERIENCE

Mechanical Design Intern, Honda of Canada Mfg. May 2025 - Aug 2026

- Mechanical design lead for multiple automation projects, owning design for robotic tooling, manufacturing fixtures, jigs, and production equipment from concept development through fabrication, installation, and validation.
- Created and maintained CATIA V5 part models, large assemblies, detailed drawings, and revision-controlled design packages used by suppliers, manufacturing, maintenance, and installation teams.
- Applied FEA, rigidity calculations, robot reach studies, payload calculations, and interference checks to validate designs and reduce technical risk before implementation.

Mechatronics Systems Engineering Co-op Student, Honda of Canada Mfg. May 2024 - Aug 2024

- Developed and delivered a course on pneumatic systems: circuit diagrams, and troubleshooting techniques
- Gained foundational experience in Mitsubishi PLC programming, supporting small-scale automation updates and troubleshooting logic-based control systems.
- Developed foundational experience with Cognex machine-vision and vision-guidance systems integrated with robotic automation.

High Ropes Team Lead, Blue Mountain Resort May 2023 - Aug 2023

- Led a team of 20 high ropes course guides, fostering a collaborative and safety-focused environment
- Acted as the primary point of contact for guest inquiries and concerns, consistently resolving issues to maintain a high level of customer satisfaction
- Maintained up-to-date knowledge of industry safety standards and ensured compliance with all relevant regulations and protocols

Sales Associate, SVP Sports 2020 - 2024

- Responsible for end of day cash tallying and balance confirmation
- Managed inventory, including daily receiving and shipping of product orders, trained new employees

EXTRACURRICULARS

Head of Technical Communications, Western Engineering Concrete Canoe Association Sept 2025 - Aug 2026

- Led the creation and management of technical documentation across multiple sub-teams, ensuring consistent communication of project workflows and producing detailed technical reports.

Big Sister Mentor, Western Woman in Engineering Club Sept 2023 - May 2025

- Mentored incoming first year engineering student providing academic guidance fostering supportive community.

Orientation Week Leader, Engineering SOPH Feb 2026 - Present

- Mentored first-year engineering students, fostering an inclusive and supportive community through transition programming, group activities, and peer guidance.

SKILLS

CAD/Design: CATIA V5, DELMIA V5, SolidWorks, AutoCAD Electrical

Programming/Controls: Java, C, C++, MATLAB/Simulink, Mitsubishi PLC

Engineering Methods & Analysis: GD&T, FEA, DFMA, Pneumatic Circuit Design, Payload Calculations, Rigidity Calculations

Professional: Technical Communication, Leadership, Project Management

SELECTED PROJECTS

HCM Grind Point Robot Automation | Mechanical Design Lead

- Used CATIA V5 to develop the robot tooling and supporting mechanical hardware for an automated grinding process, translating process requirements, robot motion, payload limits, and maintenance needs into a buildable design.
- Modelled tooling components and complete assemblies, established mounting interfaces, and performed interference and clearance checks across robot poses to confirm access to the work area and avoid collisions with surrounding equipment.
- Designed a challenging automatic tool changer arrangement, accounting for tool pickup and drop-off alignment, repeatability, pneumatic connections, payload distribution, service access, and safe retention of the grinding tool.
- Validated the tooling through robot payload calculations, reach studies, design reviews, and engineering analysis, then incorporated feedback from controls, manufacturing, and installation stakeholders.

Miniature C-Arm Fluoroscopy Simulator- Designed a 3-axis mechatronic system and 3D-modeled structural components for a scaled fluoroscopy simulator. Programmed an Arduino-based control system. Integrated accelerometer sensors for arm orientation feedback and system monitoring.

Adaptive Treat Dispenser for Wheelchair-Assisted Pet Feeding- In a small team, worked with a client and the university to design and build user-friendly treat dispenser attachment for wheelchairs to aid individuals with cerebral palsy in feeding their cats independently. The project involved a combination of mechanical design, electronics, and user interface considerations to ensure accessibility and ease of use.

Asthma Adherence aid- In the role of team facilitator, worked with a small group, the university and Trudell Medical to design a user-centered medical compliance aid to assist in increased adherence and proper technique of administering inhalers to treat asthma in young adults.

